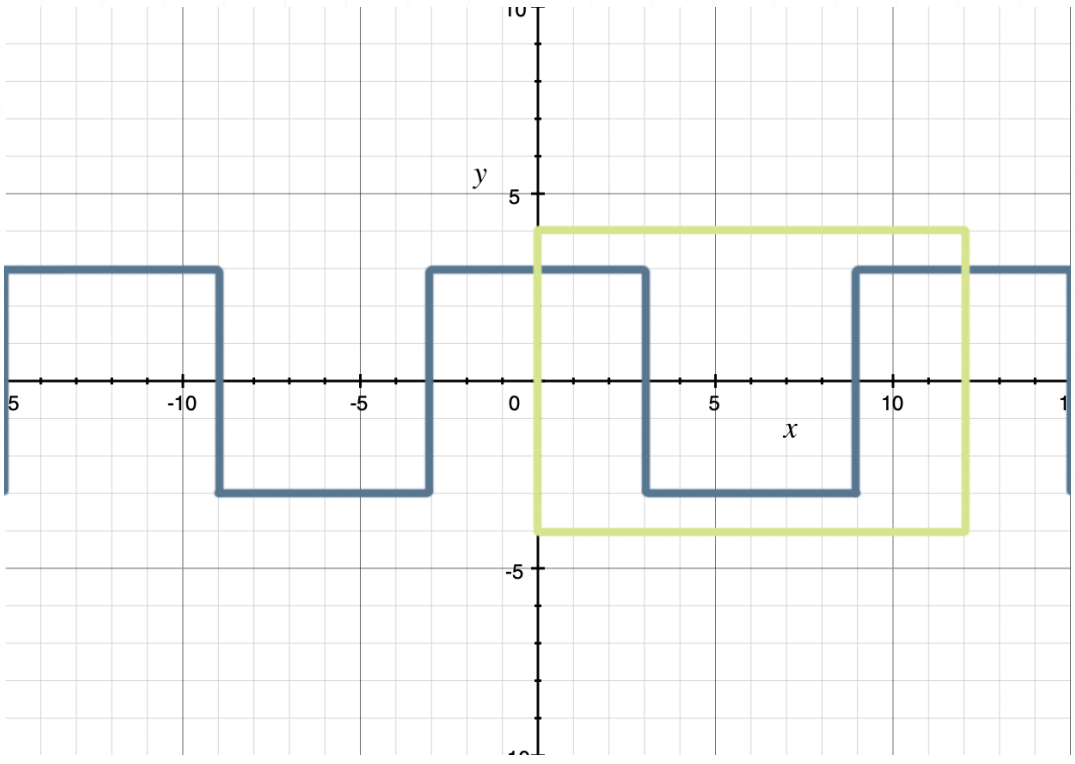
Periodic functions and periodic extensions

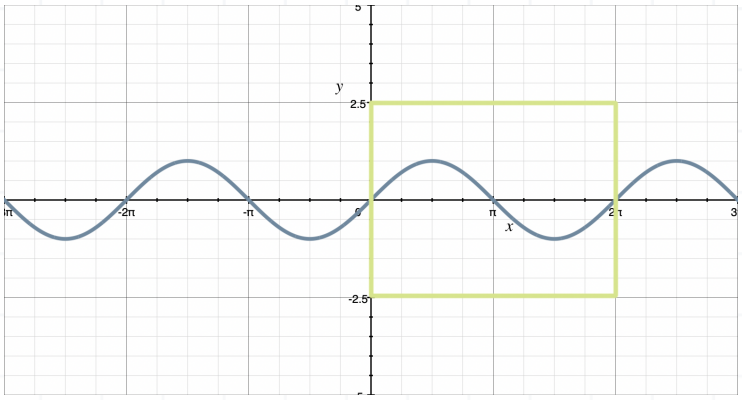
We said earlier that we can generate a Fourier series representation for any periodic function, but then we worked an example where we found the Fourier series of $f(x) = x$, which is not a periodic function.

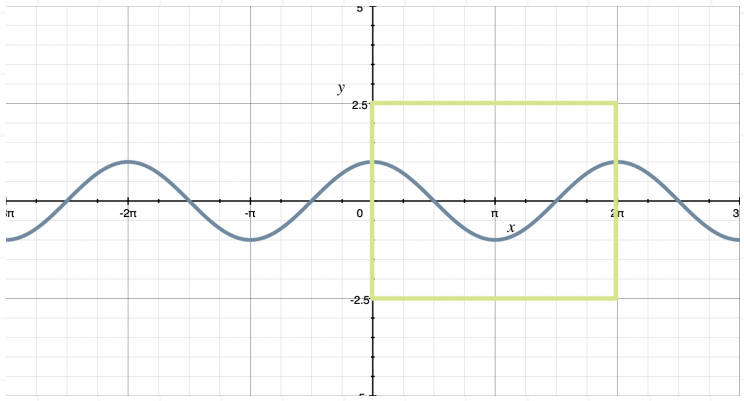
Remember that a **periodic function** repeats the same values at regular, predictable intervals. Visually, we should be able to identify the repeating pattern.

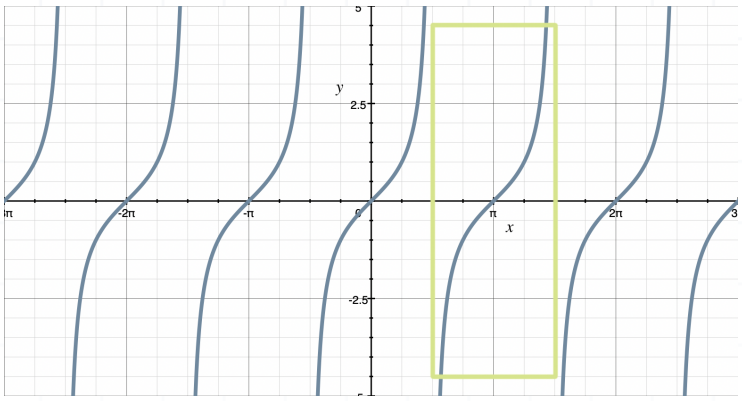


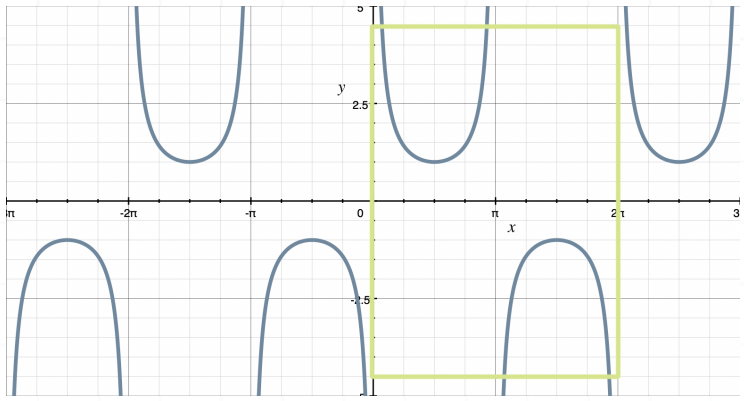
The six circular trigonometric functions (sine, cosine, tangent, cosecant, secant, and cotangent) are the periodic functions we're most familiar with, and we can see from their graphs that they're all periodic.

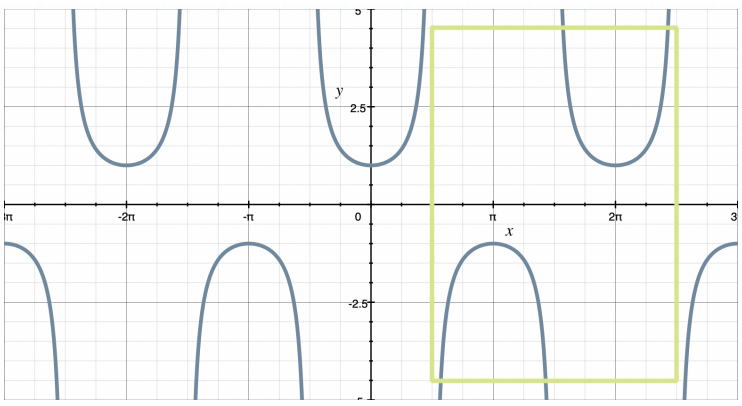
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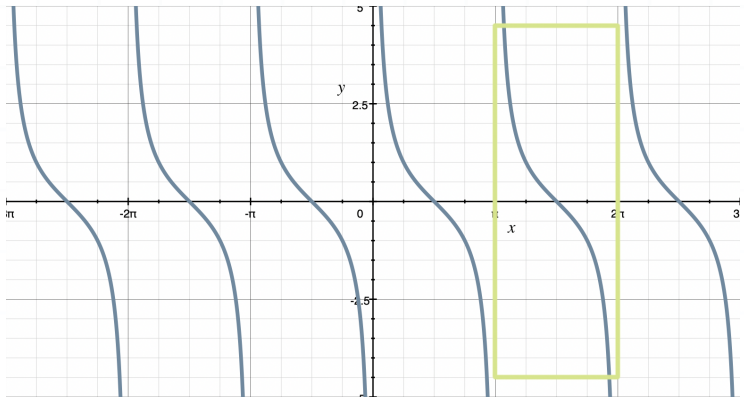












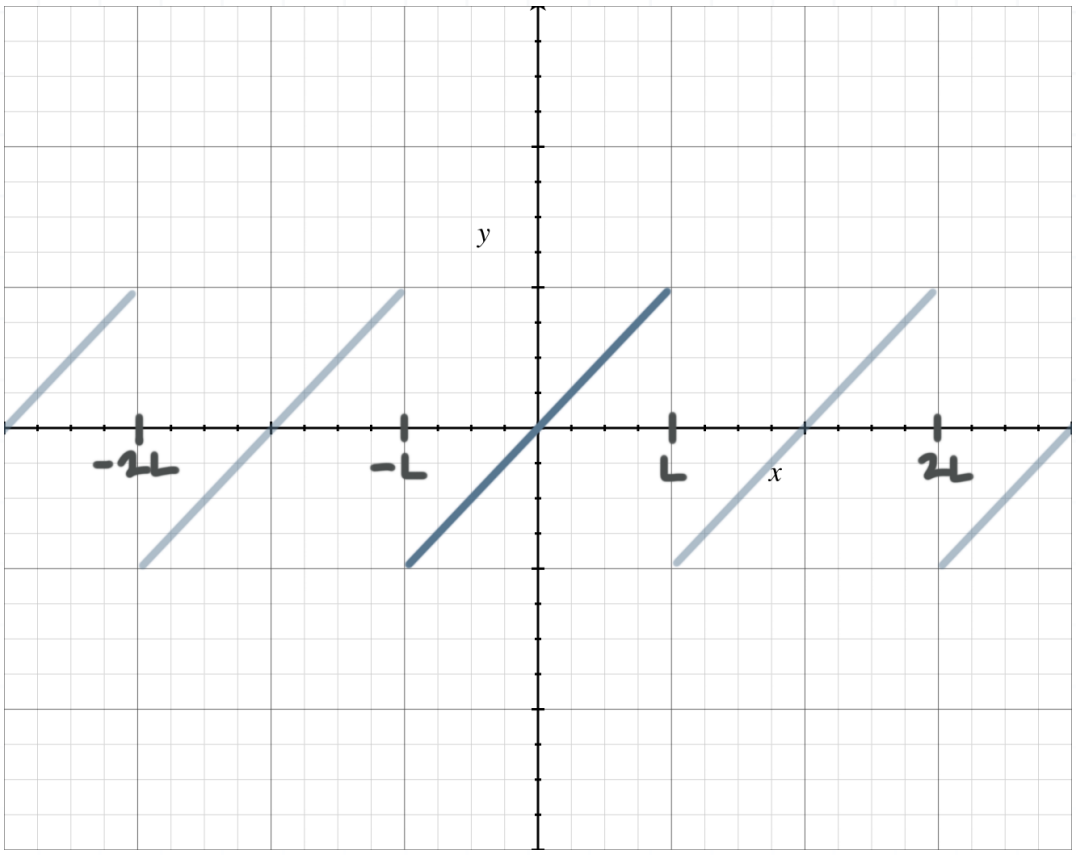
But now let's go back to the example from the last lesson, where we found the Fourier series representation of $f(x) = x$, a non-periodic function.

Periodic extensions

The only reason we were able to find a Fourier series representation of $f(x) = x$ is because we defined the representation for the interval $-L \leq x \leq L$.

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In other words, instead of representing $f(x) = x$ with a Fourier series, we actually represented the periodic extension of $f(x) = x$, which is the function we find by repeating the part of $f(x) = x$ on the interval $-L \leq x \leq L$ over and over again to the left and right.



By limiting the interval to some generic $-L \leq x \leq L$, and then repeating over and over again the part of $f(x) = x$ that we find on $-L \leq x \leq L$, we turn the non-periodic $f(x) = x$ into its periodic extension, and we can now find the Fourier series representation of the periodic extension.

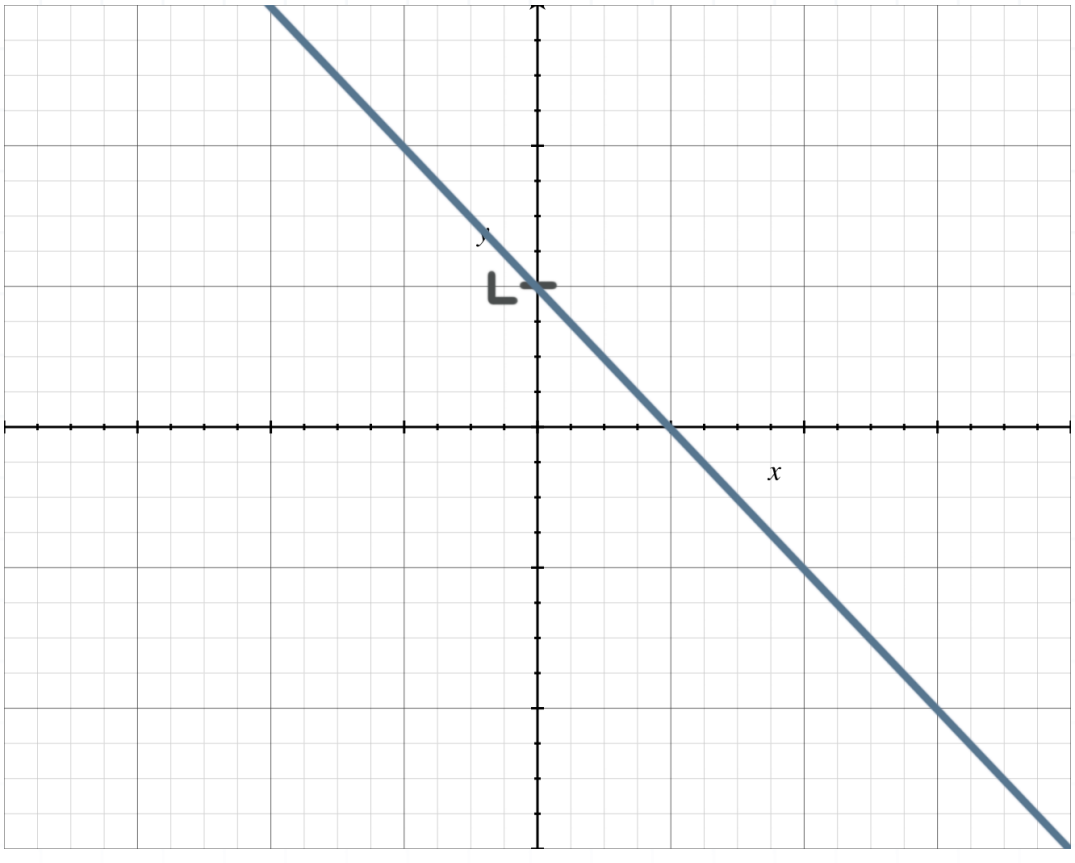
Let's do an example where we build the periodic extension of what would otherwise be a non-periodic function.

Example

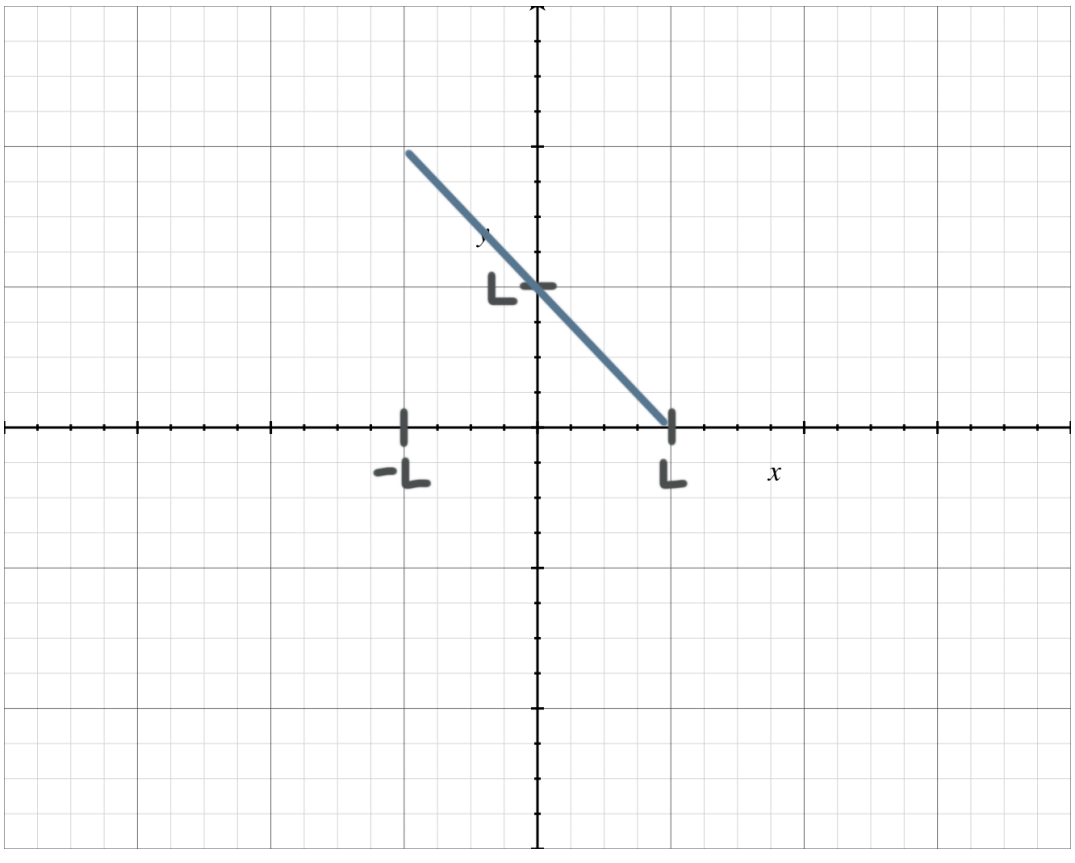
Sketch the periodic extension of $f(x) = L - x$ on the interval $-L \leq x \leq L$.

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A sketch of the function $f(x) = L - x$, given some positive value of L, could be

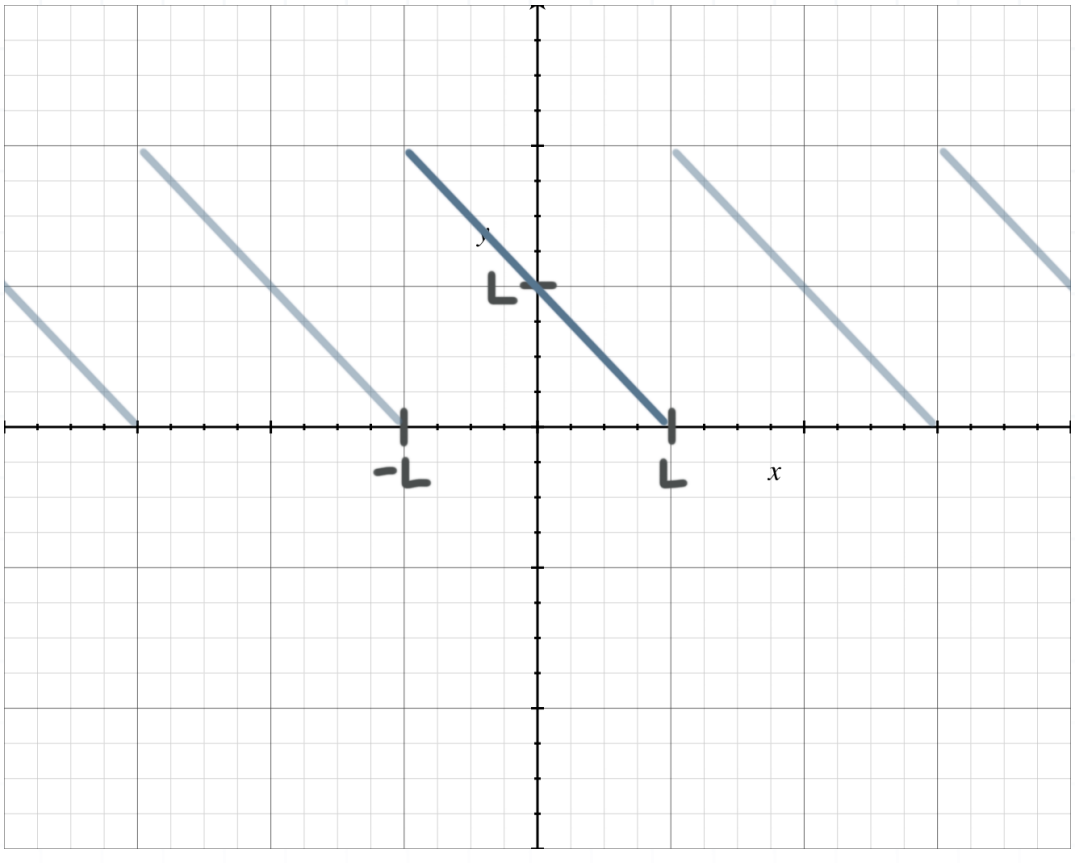


If we limit the sketch of this graph to the interval $-L \leq x \leq L$, then the section of the graph on $-L \leq x \leq L$ would be



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Now if we take this section and repeat it over and over to both the left and right, we get a sketch of the periodic extension of $f(x) = L - x$.



Even and odd extensions

It's also important that we know how to find the even extension of a function and the odd extension of a function. These particular extensions build an even or odd $g(x)$ function, from a function $f(x)$ that isn't necessarily even or odd.

Given some function $f(x)$ (which isn't necessarily even or odd), we define its **even extension** on $-L \leq x \leq L$ as

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L \leq x < 0 \end{cases}$$

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and its **odd extension** on $-L \leq x \leq L$ as

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ -f(-x) & -L \leq x < 0 \end{cases}$$

Let's do an example with the function we looked at previously, $f(x) = L - x$, so that we can compare the even and odd extensions of this function to its periodic extension.

Example

Find the even and odd extensions of $f(x) = L - x$, given some positive value of L, then sketch both functions.

The even extension of $f(x) = L - x$ is

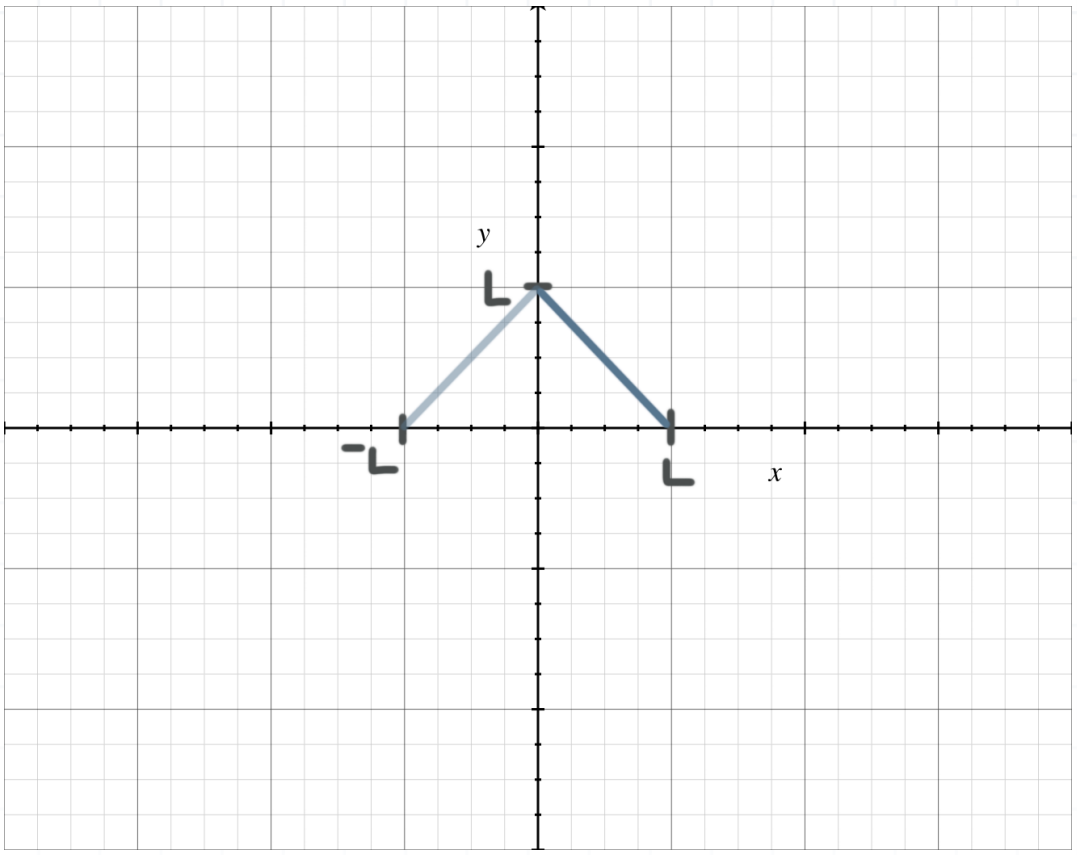
$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} L - x & 0 \leq x \leq L \\ L - (-x) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} L - x & 0 \leq x \leq L \\ L + x & -L \leq x < 0 \end{cases}$$

And the sketch of the even extension is

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The odd extension of $f(x) = L - x$ is

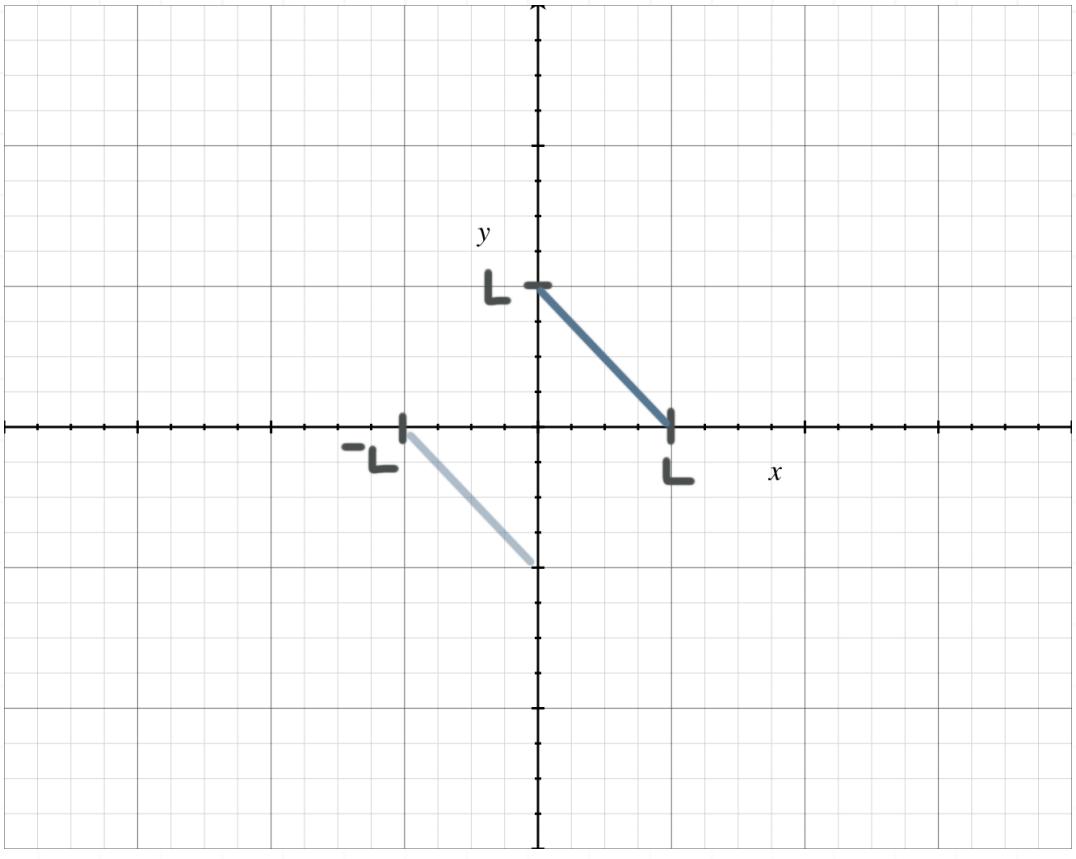
$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ -f(-x) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} L - x & 0 \le x \le L \\ -(L - (-x)) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} L - x & 0 \le x \le L \\ -L - x & -L \le x < 0 \end{cases}$$

And the sketch of the odd extension is

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